

PRISM — Technology Stack

26TS12VITV · Privacy-First AI Consent and Redaction Platform for Multimodal AI Data · SRIB-PRISM Program

Frontends

two React SPAs

Framework	React 19, React Router 7
Build	Vite 8 with @vitejs/plugin-react
Styling	Tailwind CSS 4 (@tailwindcss/vite), tailwind-merge , PostCSS + autoprefixer
UI / icons	lucide-react , qrcode.react (enrollment QR)
In-browser ML	@mediapipe/tasks-vision — live face-capture guidance in the user portal
Lint & test	oxlint; Vitest + Testing Library + jsdom (admin portal)
Dev ports	user portal :5173 · admin portal :5180

Backend

Node.js / Express API + 7 queue workers

Runtime	Node.js (ESM), Express 4
ORM & database	Prisma 5 → PostgreSQL 16 (citext , uuid-oss); ~60 models and enums
Queues	BullMQ 5 on Redis 7 (ioredis) — workers: recognition, redaction, purge, item-action, export, reaper, retention
Vector database	Qdrant v1.12 (@qdrant/js-client-rest) — face and voice galleries
Auth & RBAC	JWT (jsonwebtoken), bcrypt, refresh-token cookies, OTP; five admin roles
Security middleware	helmet, cors, express-rate-limit + rate-limit-redis, zod request validation
Cryptography	AES-256-GCM envelope encryption (MEDIA_KEK → HKDF per-subject keys); Ed25519 signing for exports and deletion certificates; SHA-256 hashing
Media handling	sharp (redaction, thumbnails, EXIF + XMP provenance stamps), multer uploads, local-disk blob store, streamed ZIP export
Email	nodemailer / Resend
Logging	pino + pino-http
Tests	node --test — 215 backend tests (unit, security, contract, e2e)

AI core

five FastAPI microservices (Python)

SERVICE	PORT	STACK
face-worker	8001	InsightFace buffalo_l (SCRFD detection + ArcFace embeddings), onnxruntime, Pillow
image-pii-worker	8002	RapidOCR-ONNX + Presidio Analyzer with custom Indian-identifier recognizers (Aadhaar, PAN, phone)
audio-worker	8003	faster-whisper (small) transcription, pyannote-audio 3.1 diarization, SpeechBrain ECAPA-VoxCeleb speaker ID, Presidio over transcripts, torch / torchaudio, soundfile — CUDA base image
text-services	8004	Presidio Analyzer + Anonymizer, spaCy (en_core_web_sm / _lg), pandas, Streamlit demo UI, OpenAI SDK

SERVICE	PORT	STACK
video-worker	8005	InsightFace <code>buffalo_l</code> (shared model volume with face-worker), opencv-headless, per-track recognition, fail-closed PII dependency

All five: Pydantic v2 + pydantic-settings, uvicorn, Python 3.10–3.12; `uv` for the audio services.

Infrastructure & tooling

Containers	Docker Compose — dev stack (<code>backend/docker-compose.yml</code> , profiles <code>localdb</code> / <code>audio</code> / <code>video</code>) and production stack (<code>deploy/compose.prod.yml</code>)
Edge	Caddy 2 — TLS termination and static portal serving in production
Orchestration	Root task runner using <code>concurrently</code> to start 10 processes; <code>scripts/setup-env.mjs</code> mints per-machine secrets
Verification	Headless Chrome driven over raw CDP (<code>Runtime.exceptionThrown</code> , <code>Network.responseReceived</code>) for UI, signup and capture-flow checks; <code>verify-export.js</code> , <code>verify-dsar.js</code> , <code>preflight.js</code>
Compliance domain	India DPDP Act — consent capture, DSAR erasure with signed certificates, retention policies, breach records, audit and access logs, DPIA

Shape of the system: a React front end and an Express/Prisma API over PostgreSQL, Redis and Qdrant, with all machine learning pushed out to five stateless FastAPI services that hold no database access and no consent logic. Redaction fails closed — if the PII detector is unreachable the pipeline defers rather than publishing unmasked media.

Generated from the repository manifests — `package.json` , `pyproject.toml` , `requirements.txt` , `schema.prisma` and the Compose files — on 1 September 2026.